

S10 methodology — every method, every number, and why

Written to be readable by someone who has not done this kind of analysis before. Each section says what we were trying to learn, what could have fooled us, what we did about it, and what the resulting number actually licenses you to claim. Numbers are as of 2026-08-02.

0. The problem, and why it is harder than it sounds

We want to know how much legislative speech is now drafted with AI help.

The naive approach — run an AI-detector over Hansard and report the percentage — fails for four independent reasons:

1. **Detectors are uncalibrated on this register.** They were validated on student essays and news. Formal prepared oratory is exactly the register they over-flag. A detector reporting "8% AI" might report the same 8% on 1998 text.
2. **There is no ground truth.** Nobody discloses. We can never build a labelled set of "AI legislative speech" from the real world.
3. **Hansard is professionally edited.** Editors normalise prose, which can erase the token-level statistical fingerprints detectors rely on.
4. **Prepared speeches were always ghost-written.** "Not written by the member" has been true for a century. Only *machine*-drafted is new, and the two are easy to conflate.

Everything below is machinery for getting a defensible number despite these.

1. Corpora

Four parliaments, chosen to vary in size, country and record format:

chamber	seats	sittings	source format	words (pre / post)
New Brunswick	49	364	bilingual two-column PDF	4.4M / 2.5M
Dáil Éireann	160	748	Akoma Ntoso XML (API)	36.7M / 18.7M
Canada Commons	338	1,097	ourcommons.ca XML	20.1M / 11.1M
UK Commons	650	1,137	ParlParse XML (rsync)	47.2M / 24.2M
US House	435	1,554	GovInfo CREC zips	22.4M / 11.3M
US Senate	100	1,554	GovInfo CREC zips	24.4M / 10.9M

Time windows are fixed by protocol: pre = 2018-01-01 to 2022-12-31, post = 2024-01-01 onward, **2023 excluded entirely** as the transition year. The 2018 floor is set by the shallowest source (NB) and every deeper archive is truncated to it, so effect sizes stay comparable (`replication_protocol.md`, Clarification 1).

1.1 What per-corpus work actually consists of

Almost all of it is *excluding text that is not the member's own English*. This is where studies quietly go wrong, so each case is documented:

- **New Brunswick** prints English and French in parallel columns, with the language *as spoken* on the left and the translation on the right. We split by word x-coordinate, language-ID each paragraph, and pair columns by vertical

overlap. Validated per speaker: anglophone members come out 94–99% original-English, francophones 20–22%. Result: 709k words of member-authored English out of 869k English words.

- **Canada** is worse, because the English edition *translates* French floor speech. `<FloorLanguage language="FR">[Translation]` marks where. **26% of scoreable "English" segments are translator-authored** and are excluded. Miss this and a quarter of your corpus is a professional translator's register.
- **Ireland** records Irish as Irish (1.0% of segments, excluded). Deputies often open in Irish and continue in English; those segments are correctly scored as majority-English.
- **UK** is monolingual, but ParlParse ships several revisions per sitting (2,937 files for 1,137 dates); we keep the one marked `latest="yes"`.
- **Chair and procedural speech** is excluded everywhere — by role attribute where the format provides one (Ireland, Canada), by name pattern otherwise.

Six New Brunswick PDFs were **truncated on the server itself** (the content-length matched the truncated bytes); recovered with pikepdf. That is why an early word count of 1.60M was wrong and the true figure is 1.89M.

1.2 Segmentation — the unit everything is measured on

1. Extract paragraphs with speaker attribution.
2. Group consecutive paragraphs into **speaker turns**.
3. Within a turn, pack paragraphs into **windows of ≤ 360 words, minimum 50**.

The 360 ceiling exists because Binoculars truncates at 512 tokens; the 50 floor because no instrument can score a one-line interjection. Every downstream number is per-window, and the *turn* is retained as the clustering unit for all bootstrap statistics (segments within one speech are not independent observations).

NB produced 37,801 scoreable windows; UK 447,210; Ireland 251,298; Canada 353,339.

2. The lexicon method — the core of the study

This is what produces the p-values. It uses **no AI detector at all**.

2.1 The instrument, and why it must be external

We count how often a fixed list of "AI-preferred" words appears, before versus after. The obvious danger: if we pick the word list ourselves after looking at the data, we will find whatever we picked. So the instrument is **Kobak, González-Márquez, Horvát & Lause (arXiv:2406.07016)** — the style-annotated subset of their excess-vocabulary list, 407 words.

Why this list is trustworthy for our purpose: - It was derived from **15 million PubMed abstracts**, not from parliaments. - It was built by an **excess-mortality-style design**: fit each word's pre-2023 frequency trend, extrapolate, flag words whose 2024 usage exceeds their own counterfactual. Nothing about it is hand-chosen for rhetoric. - It was **published before** we touched any parliamentary corpus. - Its authors hand-annotated **style vs content**; we use the style subset only, so biomedical topic words cannot leak in. (Their content list is PubMed's topic vocabulary — *lachnospiraceae*, *tocilizumab* — so it is not usable as a control here: whether biomedical jargon moved in a parliament is a fact about parliamentary topics, orthogonal to whether our style measurement is picking up AI. The frequency-and-dispersion-matched placebo already controls for in-domain churn, using words that occur in the domain.)

A second instrument — Wikipedia's *Signs of AI writing* (WikiProject AI Cleanup) — is used descriptively. It is a hand-built editorial field guide, so it is weaker evidence, but it is independent of Kobak and moves the same way, which is worth something.

2.2 The statistic, and why this one

For each word w we compute a **log fold-change**: the log ratio of its rate after to its rate before, with +0.5 smoothing so zero counts don't explode:

$$\text{logFC}(w) = \log[(k_{\text{post}}(w) + 0.5)/W_{\text{post}} \div (k_{\text{pre}}(w) + 0.5)/W_{\text{pre}}]$$

The primary statistic is the **equal-weight mean** of logFC across all 407 words — every word gets one vote.

Why equal-weight rather than pooling all the counts together? Because pooling is frequency-weighted, so a handful of common words dominate and drown out the signal. Concretely: on the full Kobak list the *pooled* ratio in NB is 1.09 ($p = 0.48$, nothing), while the equal-weight statistic is +0.379 ($p < 10^{-5}$). Same data, same words. The equal-weight version matches the hypothesis being tested — "LLMs shifted usage across their preferred vocabulary" — and, critically, it involves **no analyst choices at all**: no threshold, no subsetting. That matters for the p-hacking objection (§6).

2.3 The placebo null — why not a t-test

A textbook significance test would ask "did this rate change more than sampling noise?" With tens of millions of words, *everything* is significant by that standard. It is the wrong question.

The right question is: **is this rise unusual for words like these?** Political vocabulary churns enormously — topics arrive, phrases go in and out of fashion, a bill dominates a session. So we measure that churn instead of assuming it:

1. Build **1,000 placebo instruments**. Each is a random word set the same size as the real one, where every placebo word is drawn from the **same pre-period frequency band** (same $\log_2$ bucket) as its real counterpart. Function words, the instrument's own words, short and non-alphabetic tokens are excluded.
2. Run the identical statistic on each.
3. Those 1,000 values **are** the null distribution — they embody exactly how much matched-rarity vocabulary drifts over these same years in this same chamber, for every non-AI reason there is.
4. **p = the fraction of placebos that match or beat the real instrument.**

Two things this buys you. It needs no distributional assumptions. And the null is *not* centred on zero — placebo sets drift by -0.14 to -0.25 depending on the chamber, which is real background vocabulary churn that a naive test would have mistaken for signal.

2.4 What " $p < 0.001$ " means here

It means zero of 1,000 placebos reached the observed value, so the p-value is bounded by the resolution of the simulation, not measured. Where we ran 100,000 draws (NB) the bound is $p < 10^{-5}$, and there the real value was **4× the most extreme excursion in 100,000 draws**.

2.5 The cross-corpus normalisation (a correction)

Raw statistics are not comparable between chambers. Each corpus has its own placebo baseline, and they differ a lot. Canada makes this unmissable: its raw primary is **negative** (-0.0138) yet $p < 0.001$, because its placebo baseline is -0.231 . Canada's AI vocabulary *declined* — but far less than comparable vocabulary declined.

So the comparable effect size is **excess over the corpus's own placebo median**. Earlier drafts of this project compared raw values across chambers and were wrong to. On the corrected metric:

chamber	raw primary	excess	p
New Brunswick (<i>discovery</i>)	+0.3791	+0.554	< 0.001
Dáil Éireann	+0.0647	+0.224	< 0.001
Dáil within-speaker	+0.0722	+0.244	< 0.001
Canada Commons	-0.0138	+0.218	< 0.001
Canada within-speaker	+0.0122	+0.257	< 0.001
UK Commons	+0.1155	+0.257	< 0.001
UK within-speaker	+0.0776	+0.272	< 0.001

Fisher's method combines independent p-values: $X^2 = -2 \cdot \sum \ln(p_i)$ on 2k degrees of freedom. Over the three confirmatory chambers: $X^2 = 41.4$, $df = 6$, $p = 2.4 \times 10^{-7}$. New Brunswick is reported separately because it is the discovery corpus (§6).

2.6 Protocol v1.1 — what the effect looks like with the null repaired

Adversarial review found two defects in v1.0's null, both real, both fixed in `run_protocol_v11.py`. v1.0 is not withdrawn; both are reported, because the difference between them is itself the honest measure of how much the headline depended on the artifacts.

Fix 1 — present-in-both. v1.0 scores all 407 words, including words appearing in *neither* period. Under +0.5 smoothing each such word contributes a constant positive term, since $W_{pre} \approx 2 \cdot W_{post}$. That is an artifact of the denominators, not a fact about language.

Fix 2 — dispersion-matched placebos. v1.0 matches placebo words to instrument words on pre-period frequency alone. Corpus linguistics treats dispersion — how many distinct sittings a word appears in — as co-equal with frequency (Gries 2008; Egbert & Biber 2019), because a pool of bursty topic words decays faster between windows, depressing the placebo median and inflating every excess. v1.1 matches on both axes. **The study already knew this:** the same filter was written for the formality control (§3.1) after it returned place names, and was never ported to the placebo pool.

chamber	pre words	v1.0 excess	v1.1 excess	v1.1 p
New Brunswick (<i>discovery</i>)	4.2M	+0.554	+0.278	< 0.001
UK Commons	47.2M	+0.257	+0.121	< 0.001
Dáil Éireann	36.7M	+0.224	+0.080	0.002
US House	22.4M	+0.299	+0.104	< 0.001
Canada Commons	20.1M	+0.218	+0.029	0.132 — n.s.
US Senate	24.4M	+0.151	-0.036	0.910 — n.s.

Fisher over the five confirmatory chambers: $X^2 = 44.3$, $df = 10$, $p = 2.9 \times 10^{-6}$. The combined result stands. **Two of five chambers do not reach significance individually.** Effect sizes roughly halve against v1.0. Any claim of "consistent magnitude across parliaments" is dead; the defensible range among confirmatory chambers is -0.03 to +0.12.

The US is now the **complete record** for both windows — 1,554 sitting days, 71M words — so both US rows are adequately powered and directly comparable to the other chambers. (The earlier 330-day sample gave ~3M words per chamber and inflated excesses, because sampling thins per-word counts and drags the placebo baseline down; those numbers are superseded.)

The two nulls are now real nulls, and one of them is the most useful comparison in the study. The US House (+0.104, $p < 0.001$) and the US Senate (−0.036, $p = 0.910$) sit in the same country, the same era, the same published Record, under the same national media environment — and differ completely. Whatever separates them is institutional, not national. That is a within-country contrast no cross-country comparison can give, and it is the sharpest evidence in the study that the effect is not some artifact of Anglophone parliamentary drift.

The Senate null is not the instrument failing to work there — its style arm behaves like the others, it simply does not move.

2.6b THE NULL DOES NOT CONTROL FOR TREND — read this before quoting any p-value

Every significance claim in this section is currently unsupported. The frequency-and-dispersion-matched placebo has now been run on windows where the effect *cannot* exist, and it fails.

Take window pairs entirely before ChatGPT and run the identical v1.1 protocol:

chamber	REAL 2018–22 vs 2024–26	2018–19 vs 2021–22	2018 vs 2020	2019 vs 2021	estimator check
UK Commons	+0.120	+0.123	+0.130	+0.165	+0.011 ($p=0.26$)
Dáil Éireann	+0.083	+0.112	+0.102	+0.094	+0.030 ($p=0.06$)
Canada Commons	+0.033 (<i>n.s.</i>)	+0.037	+0.072	+0.087	+0.008 ($p=0.36$)
New Brunswick	+0.236	+0.137	+0.204	+0.060	+0.066 ($p=0.01$)

Pre-LLM windows return excesses **as large as or larger than the real comparison**, at $p < 0.001$, in three of four chambers.

The estimator is not the problem. Splitting the *same* period into interleaved halves by alternating sitting date — identical time span, differing only by sampling — returns ~0 (+0.011, +0.030, +0.008). So the machinery is near-unbiased and the in-time placebos are detecting a **real secular trend**: this vocabulary has been rising in parliamentary English since at least 2018.

Why the null cannot catch it. A frequency-and-dispersion-matched placebo controls a word's *level* and its *burstiness*. It does not control *trend* — matched words do not share the instrument's drift. So "the instrument rose more than matched words did" is exactly what a already-rising vocabulary produces, with or without LLMs.

The diagnosis: we imported Kobak's word list but not Kobak's estimator, and the estimator was doing the work. Their design fits each word's own pre-trend and measures the *break* from it — $q = p_{-2} + 2 \cdot \max\{p_{-2} - p_{-3}, 0\}$, floored so it can never predict a decline — which is a trend control by construction. We replaced it with a two-window level contrast plus a placebo, and a placebo cannot recover what the extrapolation was doing.

What this invalidates: every lexicon p-value, the Fisher combination, and the cross-chamber effect-size comparisons. New Brunswick is the only chamber whose real effect clearly exceeds its own in-time placebos, and it is the discovery corpus.

What it leaves untouched: the prevalence arm (§5 — Pangram verdicts and specificity calibration involve no time-series contrast), the base-versus- instruct result (§6.1 — a property of model outputs, not of Hansard over time), and the detector-side permeation evidence (§5.2a).

The fix is a rebuild, not a patch: replace the two-window primary with a trend-aware counterfactual in the manner of Kobak's extrapolation, and re-run every chamber. Until that exists, the tables above and in §2.5–2.6 should be read as descriptive, not inferential.

2.6a The per-year series — the pooled ratio was hiding two things

Pooling 2024–2026 into one number assumes the effect is stable inside the post window. It is not. Each post year scored against the same 2018–22 baseline, under v1.1, with its own placebo null:

chamber	2024	2025	2026
New Brunswick	+0.112	+0.266	+0.105
Dáil Éireann	+0.050	+0.106	+0.096
Canada Commons	−0.019 (n.s.)	+0.068	+0.058
UK Commons	+0.102	+0.127	+0.036 (n.s.)
US House	+0.159	+0.034 (n.s.)	+0.038 (n.s.)
US Senate	+0.051	−0.062	−0.054

First: Canada's pooled null was a mixture artifact, exactly as the review said. Canada is 49% 2024 by words, and 2024 is its only null year. Scored per year it is positive and significant in **both 2025 (+0.068) and 2026 (+0.058)**. The pooled Canada figure should not be read as "no effect in Canada" — it is "no effect in the half of Canada's window that is 2024".

Second, and more awkward: the series is not a rising ramp. Four of six chambers **peak and then decline**, and the US Senate goes negative. A narrative of steady acceleration — which an earlier progress report asserted — is not supported.

The most likely explanation is that the instrument is decaying, not the behaviour. Kobak's list was derived from 2024 data, and Geng & Trotta show individual markers collapsing once they become notorious (*delve* fell sharply through 2024 while other ChatGPT-favoured words kept rising). If writers and models both drifted off the 2024 tells, the lexicon signal falls while actual use does not.

The detector evidence supports that reading, because the two instruments **diverge in exactly the direction instrument-decay predicts**:

chamber	lexicon 2025 → 2026	Pangram AI rate 2025 → 2026
UK Commons	+0.127 → +0.036	1.0% → 1.6%
Dáil Éireann	+0.106 → +0.096	3.3% → 4.4%
Canada Commons	+0.068 → +0.058	13.8% → 13.7%
New Brunswick	+0.266 → +0.105	4.3% → 13.6%

New Brunswick is the clearest case: its lexicon excess more than halves while its measured prevalence **triples**. A word list frozen in 2024 cannot be assumed to track behaviour in 2026, and any claim about the *trajectory* of AI use should rest on the calibrated detector, not on the lexicon. The lexicon's job is establishing that a shift happened at all, against a defensible null — not measuring how fast it is growing.

2.7 Does the lexicon predict an independent detector?

The obvious challenge to any word-list instrument is that it might be measuring nothing in particular. The test that bears on this is not whether some other word list moved — it is whether **lexicon density predicts what a calibrated detector says, segment by segment**.

Two detectors are available and neither has any connection to Kobak's list: Pangram (calibrated in-domain, 423/423 specificity, but expensive so only ~120 unbiased segments) and the Opus screen (AUC 0.951 against Pangram, run over all 37,801 NB segments).

The list is specific. Per-segment density predicts the Opus score at $p = +0.252$, while **frequency-and-dispersion-matched random word sets predict essentially nothing**:

era	Kobak list	matched random sets	placebos beating Kobak
2025–26 (n = 7,218)	+0.252	+0.024 (−0.072 to +0.116)	0 of 30
pre-2023 (n = 20,329)	+0.250	+0.004 (−0.103 to +0.121)	0 of 30

So this is not "passages with more mid-frequency abstract words look AI-ish to everything." The list carries particular information an independent detector recognises. Pangram agrees: within the unbiased random stratum, density separates its verdicts at AUC 0.680 ($p = 0.031$), matching the Opus figure of 0.691 on 60× the data.

But the association is the same size before AI existed, and that is the important qualification. What the two instruments agree about is a **register** — a recognisable way of writing that predates these tools and that LLMs amplified rather than invented. Their agreement validates Kobak's list as an instrument *for that register*; it does **not** establish that either detects AI at the segment level. Per-segment claims of the form "this speech is machine-drafted because it is lexicon-dense" are not supported.

The levels are what carry the signal, and they move very differently:

	pre-2023	2025–26	change
lexicon density (per 1,000 words)	32.13	35.98	+12%
Opus AI score	8.18	18.03	+120%

And the detector's rise is not the lexicon's rise. Matching segments on *absolute* density in 2.0-wide calipers — so the two eras agree on mean density to two decimal places within each bin — the score shift persists at every level, +5.3 at the lowest to +17.3 at the highest:

	pre-2023	2025–26	shift
density-matched weighted mean score	8.68	18.86	+10.18
unmatched overall	8.18	18.03	+9.85

Conditioning on density **does not shrink the gap; it slightly widens it**. If the two instruments were measuring one thing, matching on density would absorb the detector's rise. It absorbs none of it, so the lexicon's +12% explains essentially none of the detector's +120%.

(An earlier version of this test binned by density *decile*. That is not an adequate control: the density distribution shifted right, so within any decile the post-period segments sit higher in the bin than the pre-period ones, and part of the apparent shift is residual density. Absolute-density calipers fix it, and the result survives.)

Three things follow. The **lexicon understates the change** (12% against 120%), so its effect sizes are conservative. The two arms measure **different quantities** — register shift versus drafting prevalence — and should be reported separately rather than treated as cross-validating. And the detector sees structural change the 407 words miss, which is why the prevalence arm is not redundant with the lexicon arm.

Caveat: the Opus screen's threshold is well calibrated (0.0% flag rate across 20,329 pre-2023 segments, Pangram-confirmed 243/243 human), but its continuous score is not anchored the same way, and Opus was trained on data that includes AI-influenced recent text. The decile shifts are large enough that this is unlikely to explain them, but it cannot be ruled out here.

3. Controls — what each one kills

A result is only as good as the alternative explanations it excludes.

control	the alternative it kills	result
Pre-2023 window	"detectors/lexicons just flag formal oratory"	every chamber's own pre-AI baseline
Placebo null	"all vocabulary drifts like this"	every placebo draw negative, instrument positive, in all four chambers
Formal-register control	"this is just pomposity inflation"	NB 0.808, Ireland 0.990, Canada 0.987, UK 0.998
Data-defined formality axis	"you hand-picked a control that passes"	instrument leads it everywhere (§3.1)
Within-speaker restriction	"the chamber's membership changed"	survives in all three chambers that held elections
Absent-word check	"smoothing manufactures the gap"	present-only restriction <i>widens</i> NB/UK gap
Translation exclusion	"you're measuring translators"	NB columns, Canada FloorLanguage, Ireland language-ID
Speech-clustered bootstrap	"a few speeches drive it"	resample whole turns, not words

3.0 One assumption worth stating, because we measure its violation

Most causal designs assume **no interference** (SUTVA): one unit's treatment does not affect another unit's outcome. Here that would mean AI-assisted writing does not change how *non-AI* text is written. **We know it does** — that is the permeation result in §5.2, and it is the study's most interesting finding rather than a nuisance.

The consequence is worth writing down because it runs in our favour. The placebo pool has no semantic exclusion, and the formality control is built to be maximally register-similar — so both are partly *treated*. Contaminated controls make the comparison too conservative, biasing every reported effect **toward the null**. Whatever the true shift is, the numbers here are a floor, not a ceiling.

3.1 The data-defined formality control (worth understanding)

The hand-picked register list invites the objection "you chose words you knew would behave". So we built one from data instead:

```
formality score(w) = log[ rate(w | long pre-period speeches ≥300w)
                        ÷ rate(w | short interventions ≤120w) ]
```

Long speeches are overwhelmingly prepared, short interventions overwhelmingly spontaneous, so words scoring high mark the prepared register. Take the top scorers, frequency-match them bucket-for-bucket to the instrument, require zero overlap with it, and use **pre-period data only** so nothing about the outcome leaks in.

The first attempt failed, instructively. It returned *hessle*, *mcdonagh*, *broxbourne*, *smokers*, *circuses* — place names and bill topics. Long speeches differ from short ones by *topic*, not just register. Two filters fixed it: a dispersion floor (a word must appear in at least as many sittings as the median instrument word) and a proper-noun filter (excluded if capitalised mid-sentence >25% of the time). The axis then returns real register vocabulary: *amends*, *movingly*, *poignant*, *deficiencies*, *discourse*, *practicable*.

A third filter was needed and the review found it (F7). The candidate floor was a pre-period count of 200, which sits above log2 bucket 7 — but 227 of the instrument's 407 words live in buckets 0–3, so those quotas could never fill. The

realised control was 67 words in New Brunswick against the instrument's 407, and was therefore systematically **more common** than the thing it controls — reintroducing the exact flaw it exists to remove. Lowering the floor to 20 and borrowing from neighbouring buckets when one still cannot fill:

chamber	control words filled	instrument – control
New Brunswick	407 / 407	+0.392
UK Commons	362 / 407	+0.092
Dáil Éireann	371 / 407	+0.077
Canada Commons	381 / 407	+0.074

The instrument leads the control in every chamber, as before, and the gaps barely move — so the earlier numbers were not *wrong*, they were under-evidenced. What changed is that the control is now actually matched.

One honest weakness remains. The control vocabulary looks like register in the three large chambers (*whilst, amends, movingly, poignant; consisting, pleaded, nowadays; thereafter, ramifications, illustrate*) but topical in New Brunswick (*drainage, plantations, asphalt*). NB has only 169 pre-period sittings, so its dispersion filter is coarse — the fix that worked at scale degrades on the shallowest corpus, which is another reason NB is quoted as the discovery corpus rather than as evidence.

4. Detectors — three tiers, and why the first two failed

4.1 Statistical / zero-shot (all failed)

Eight instruments: three **Binoculars** pairs (Falcon-7B, Qwen3-1.7B, Qwen3-8B), **Fast-DetectGPT**, **log-rank ratio**, and three supervised classifiers (HC3-RoBERTa, RADAR, the GPT-2-era OpenAI detector). Each calibrated on the 2019 control so its threshold means the same thing.

Binoculars scores the ratio of a text's perplexity under one model to its cross-perplexity under a closely related model — machine text is "unsurprising to a machine" in a specific way. It is a good method. It found nothing: **2025–26 flag rates sat below the pre-ChatGPT false-positive floor** (2.85% vs 5.64%). Fast-DetectGPT and the classifiers agreed.

That null is about the detectors, not the text, and real data shows it. The obvious worry is that "eight detectors found nothing" might just mean zero-shot detectors do not work on transcribed parliamentary speech. It does not, because *other* instruments find plenty in the very same corpus: Pangram individually confirmed **576 of 643** tail segments as AI or Mixed (89.6%), and a frontier LLM screen separates its verdicts at AUC 0.951 while open-weight models manage 0.55–0.72 (§4.2).

So the same segments that eight statistical detectors rate unremarkable are adjudicated AI by a calibrated commercial classifier and recovered by a frontier model. The null is a property of the *method* — perplexity-based detection — not evidence that the record is clean. That is the scissors: the vocabulary channel and the commercial/frontier detectors both move, while token-level statistics do not.

This argument previously rested on measuring sensitivity against 40 synthetic speeches generated by Mistral-7B-Instruct. That corpus has been removed from the study entirely: it was a single 7B open-weight model nobody's office uses, too small to carry variance (39 of its 89 usable words occurred exactly once), and it is not needed — the case above is made on real adjudicated text rather than on our guess at what machine legislative speech looks like.

4.2 Prompt-based detection, and the capability finding

Same blinded pool, same question, six configurations:

detector	AUC vs Pangram
Claude Opus 5 (low effort)	0.951
Claude Fable 5 (high effort)	0.936
Qwen3-32B (thinking)	0.721
gpt-oss-120b (high reasoning)	0.672
gpt-oss-120b (low reasoning)	0.657
Qwen3-32B (no thinking)	0.549

Reasoning effort helps the open models somewhat but never closes a ~0.25 AUC gap. **Detecting edited AI in this register is frontier-capability-bound** — which retro-explains why the statistical detectors failed, and is itself a finding.

4.3 Calibration — the part that makes a percentage meaningful

A detector's output is worthless without knowing its error rates. Two parameters:

- **Specificity (Sp)** = P(not flagged | actually human). **Measured here**, in-domain, on pre-2023 segments that predate the tools.
- **Sensitivity (Se)** = P(flagged | actually AI). **Taken from the published literature**, because no study — including this one — can measure it for the text that matters.

Specificity: 423 pre-LLM segments, zero false positives. New Brunswick 243/243, then 60/60 independently in each of Ireland, Canada and the UK. The NB figure includes the three pre-2023 segments an LLM screen rated *most* AI-like, all of which came back Human — a specificity test chosen adversarially rather than at random. This is the half of the calibration that had to be done in-domain, because a detector's false-positive rate on formal parliamentary oratory is exactly what no vendor benchmark covers.

Sensitivity: cite, don't generate. Pangram's catch rate on machine text is far better established publicly than anything 40 speeches of ours could show: 99.3% on the RAID benchmark across domains (Dugan et al., COLING shared task); never below 99.8% in independent testing at Chicago Booth; the only one of four tools to detect reliably in a peer-reviewed Vrije Universiteit Brussel evaluation; 1.4% false-negative rate self-reported in the Pangram technical report (Emi & Spero, arXiv:2402.14873). Pangram also trains against 19 humanizer tools as augmentation, which bounds how badly light editing should hurt.

An earlier version estimated Se from 40 synthetic speeches generated by Mistral-7B-Instruct (Se = 40/40). That corpus has been removed from the study. It was weaker evidence than the published evaluations it duplicated, and it rested on a single 7B open-weight model nobody's office uses — the wrong generator for asking what share of real speech is machine-drafted. Nothing now depends on it.

What nobody has measured, and why it cannot be fixed by generating more text. Every published figure above is for machine text as emitted, or perturbed by automated humanizers. The class this study argues actually dominates Hansard is different: drafted by an aide with a model, revised by the member, then normalised by a professional Hansard editor. To measure Se on *that*, you would have to generate it — which means deciding what a member's edit looks like. That is an unvalidatable modelling assumption, and it would yield a confident-looking sensitivity figure resting entirely on a guess. Reporting the floor honestly is better than manufacturing an interval.

Rogan–Gladen correction. An imperfect test's observed flag rate π relates to true prevalence τ by

$$\pi = \tau \cdot \text{Se} + (1 - \tau)(1 - \text{Sp}) \quad \text{so} \quad \tau^{\wedge} = (\pi^{\wedge} + \text{Sp} - 1) / (\text{Se} + \text{Sp} - 1)$$

↑ caught ↑ falsely flagged

Because **Sp = 1 is measured**, the false-positive term vanishes and this collapses to

$$\hat{\tau} = \hat{\pi} / \text{Se} \quad \text{and since } \text{Se} \leq 1 \text{ always, } \hat{\tau} \geq \hat{\pi}$$

So the correction contributes no number — it tells us the *shape* of the remaining uncertainty. It is **one-sided and upward**, with $1/\text{Se}$ as the unknown multiplier. Every observed rate in §5 is therefore a floor, and the only thing that could move it is evidence about sensitivity on edited in-register text, which does not exist.

The denominator ($\text{Se} + \text{Sp} - 1$) is how much better than chance the test is; as it approaches zero the estimate explodes, which is why a mediocre detector cannot produce a prevalence figure at any sample size. Ours is dominated by the measured $\text{Sp} = 1$, which is the part that makes the floor meaningful.

4.4 Two-phase sampling (why the design is affordable)

Adjudicating a whole corpus with a paid detector is unaffordable. Two-phase sampling solves it: a cheap instrument scores everything and defines strata, then the expensive instrument is spent only where it is informative. **The cheap instrument does not need to be accurate — only correlated with truth — because stratum weights are known exactly from the full corpus.**

In practice a lean Claude Opus screen (low effort, detection-only prompt) scored all 37,801 NB segments for ~17M tokens, at AUC 0.951 against Pangram. Its precision is monotone in score — 100% at 90+, 83% at 50–59 — which is what a well-behaved stratifier looks like.

5. The prevalence numbers

- **Design-based random sample:** 120 segments drawn 12-per-decile of the **Falcon/Binoculars** score. Result **9 AI + 1 Mixed of 120: 7.5% AI** [4.0–13.6], **8.3% AI-or-Mixed** [4.6–14.6]. Report which of the two you mean — Pangram returns three verdicts, not two, and the AI-only and AI-or-Mixed rates diverge sharply in other chambers (Canada: 17.5% vs 26.7%).

Two corrections a reviewer would otherwise make first (review F4).

The reference population is five sitting months, not the corpus. The sampling frame is 3,998 segments — 585 from 2019 plus Oct/Nov/Dec 2025 and Mar/May 2026 — which is **10.6% of the 37,801-segment corpus**, omits all of 2024 and January–September 2025, and sits on the steepest part of the ramp. So the defensible statement is "**7.5% of New Brunswick segments in sittings from October 2025 to May 2026**", not a corpus rate. The other three chambers use a different window again (2025-01-01 onward), which is why §5.0b's table is labelled 2025–26.

The stratifier was uninformative. The deciles are of the Falcon score — the Binoculars detector §4.1 reports as having found nothing — not of the Opus screen. Realised AI counts by decile were 0,1,1,1,1,0,1,1,1,2: flat. The design therefore **degenerated to an effective simple random sample**. That is harmless for the point estimate under equal allocation across equal-sized deciles, but it must be stated, and §4.4's "precision is monotone in score" describes the Opus screen on a different pool. - **By year:** 0% (2019–22, $n=240$) → 4.0% (2023) → 4.0% (2024) → 4.3% (2025) → **13.6% (2026)**. - **Full tail adjudication:** all 643 segments scoring ≥ 50 submitted individually → **461 AI, 115 Mixed, 67 Human (89.6% confirmed)**. - **Corpus-wide screen:** pre-2023 flag rate **0.0%** across 20,329 segments; 0.5% (2023) → 1.2% (2024) → 7.0% (2025) → 11.0% (2026).

Two numbers that must not be conflated. 576 confirmed segments over 37,801 is a **1.5% audited floor** — it only counts what was individually verified. 7.5% is the **estimated prevalence** from random sampling. They measure different things.

And 7.5% is a floor, not a point estimate (review F5). Since $\text{Sp} = 1$ is measured, $\hat{\tau} = \hat{\pi}/\text{Se}$ (§4.3), so the only thing standing between the observed rate and the true one is sensitivity on the text that actually dominates: **edited AI**, in

parliamentary register. Nobody has measured that — not this study, not any published evaluation. What it would do to the number:

Se on edited in-register AI	implied prevalence
1.00	7.5%
0.75	10.0%
0.50	15.0%
0.30	25.0%

Two things bound how pessimistic to be. Against us: Dugan et al. (RAID) show detector performance collapsing under light edits across eight detectors, and Liang et al.'s proofreading arm found lightly-edited text yields only a slight, non-significant rise in their estimator — both say editing hurts. For us: those results are for detectors that do **not** train on humanized text, whereas Pangram trains against 19 humanizer tools as augmentation, so the most pessimistic rows are probably too pessimistic.

This gap is not closeable by more work, which is why we stopped. Measuring Se on edited in-register AI means generating it, which means deciding what a member's edit looks like — an assumption nothing can validate. The result would be a precise-looking interval resting on a guess. The defensible statement is the one-sided one: **at least 7.5% of New Brunswick segments in sittings from October 2025 to May 2026, with 1/Se as a stated unknown multiplier,** and every correction running toward more AI, never less.

5.0a Segment length is a confound, and it bit us

The three confirmatory chambers were first sampled with a **120-word floor** (so that Pangram had enough text to work with). That floor is not innocent: AI-flag rate rises steeply with segment length. In Canada, 11.1% of 120–199-word segments were flagged versus 22.6% of 280–360-word ones, and the excluded 50–119-word band came back at **2.5%** against the long band's 17.5%. Since that band is **45.8% of Canadian segments**, the unweighted long-band figure overstates the corpus rate by roughly 1.7×.

The fix is to measure the excluded band rather than extrapolate, then combine with weights taken from the **full corpus** (not the sample), which is what makes the result design-based:

chamber	long band	short band	segment-weighted	word-weighted
Canada	17.5%	2.5%	10.6% ± 4.3	14.0% ± 5.3

Two weightings because they answer different questions: segment-weighted is "what share of *turns* is AI-flagged", word-weighted is "what share of the *record* is". Quote the second only if the claim is about text volume.

New Brunswick needed no such correction — its sample was 16% short segments against 13.9% in its pool, so it was already representative. **This is why NB (7.5%) and Canada (10.6%) are statistically indistinguishable, and why the uncorrected comparison — 7.5% vs 17.5% — would have been an artifact of the sampling frame rather than a fact about Canada.**

5.0b Four chambers, corrected — and they genuinely differ

The three confirmatory chambers were sampled **uniformly at random** from their post-window pools, with no screen stratification, so unlike the NB pilot they need no weighting assumptions at all. Every one carries its own specificity control drawn the same way from pre-2023.

chamber	seats	specificity	long band	short band	AI, corrected	AI+Mixed
UK Commons	650	60/60	1.7%	0/40	0.8% ± 1.0	2.3% ± 1.8
Dáil Éireann	160	60/60	5.0%	0/40	4.3% ± 3.3	9.9% ± 4.9
New Brunswick	49	243/243	8.9%	0/19	7.5% [4.0–13.6]	8.3% [4.6–14.6]
Canada Commons	338	60/60	17.5%	2.5%	10.6% ± 4.3	15.6% ± 4.8

Two things to take from this.

Specificity replicates. 423 pre-2023 segments across four parliaments, zero false positives. The NB pilot's perfect specificity was not a fluke of one corpus, which is what pins down the specificity half of the Rogan–Gladen correction rather than a fudge.

The chambers really do differ. A homogeneity test on the long band, the one stratum sampled identically everywhere, gives $X^2 = 22.2$, $df = 3$, $p = 6.0 \times 10^{-5}$ — a tenfold spread from UK 1.7% to Canada 17.5% that sampling noise does not explain.

And prevalence does not rank chambers the way the lexicon does. The UK has among the largest lexicon excesses (+0.257) and the *lowest* measured prevalence; New Brunswick has by far the largest excess (+0.554) and a middling prevalence. So the two instruments are not two measurements of one quantity, and the paper must not treat them as interchangeable. Three candidate explanations, none yet tested:

1. **Editorial policy.** UK Hansard is the most heavily edited of the four — it reports substantially what was said, with repetition and error removed. Heavy editing is exactly what destroys detector signal while leaving word choice intact. This is the scissors (§5.1) operating at chamber level, and it predicts the UK's specific pattern.
2. **Register composition.** The UK has the most short-intervention-heavy record of the four (56,380 short vs 46,715 long segments), and short segments are both harder to detect and less often drafted.
3. **Genuine behavioural difference** in how members prepare.

Distinguishing these needs an editing-intensity measure the study does not yet have. Until then the honest statement is that the **lexicon measures register drift, including ambient permeation into human writing (§5.2), while the detector measures drafting** — and that they diverge.

5.1 The scissors, and what it implies

Vocabulary rises while token statistics do not. That combination is the signature of **edited or assimilated** AI: editing destroys the token-level fingerprint but not the word choice. It also explains why a study designed around detector flag rates alone would have concluded, wrongly, that nothing was happening.

5.2 Permeation into human-written text

If AI register is spreading into prose people write themselves, the rise should appear even in text a calibrated detector calls human. Testing on segments the screen rates *least* AI-like (score <20, identical selection rule both eras):

era	words	Wikipedia-signs/100k	rare-Kobak/100k [95% CI]
pre-2023	4.08M	11.4	213.7 [209.3, 218.2]
2025–26	1.06M	16.1	264.8 [255.1, 274.7]

Non-overlapping intervals, 1.24–1.41× higher. **Part of the corpus-level rise is ambient** — the machine register colonising human writing — not only direct drafting.

That test used a screen score as a human proxy, which is vulnerable to an obvious objection: if members run drafts through a humanizer, "the detector says human" means the detector lost, not that a person wrote it. The permeation case therefore cannot rest on any detector, and the two tests below are built so that it doesn't.

5.2a Dose-response by speech format — the strongest evidence

Drafting and permeation make opposite predictions about *where* the shift should sit. Drafting enters where drafting is easy and worthwhile: long prepared statements. Permeation enters wherever people produce language, including short spontaneous interventions nobody drafts with a model.

Segments split at 120 words, the frozen protocol run on each band separately, and the same blinded Opus screen (n = 400 per cell) run on both:

chamber	lexicon excess, short	lexicon excess, long	detected AI, short	detected AI, long
Dáil Éireann	+0.378	+0.214	0.0%	0.2%
Canada Commons	+0.312	+0.221	0.2%	2.8%
UK Commons	+0.299	+0.267	0.0%	0.8%

The two columns run in opposite directions in all three chambers. Detected drafting concentrates in the long band; the vocabulary shift concentrates in the short band. If drafting caused the vocabulary shift these would track each other.

Two things make this hold up. First, the obvious objection — that the screen just scores short text lower — is settled by the pre-2023 cells: **0 flags in 2400 pre-LLM segments**, so there is no baseline short/long gap to subtract. Second, the argument needs only the detector's *ordering* across bands, not its absolute accuracy, so a humanizer that defeats detection does not rescue the drafting story unless it is being applied to seventy-word Question Period supplementaries.

The screen also sees the permeation directly, below its own flag threshold: mean score rises significantly pre → post in **every** short-band cell — Ireland +1.18 [0.65, 1.75], Canada +0.95 [0.42, 1.54], UK +2.31 [1.84, 2.80] — while almost nothing crosses the threshold. Text reading slightly more machine-like without being machine-written is exactly what permeation looks like.

5.2b Speaker distribution — is it everyone, or a few adopters?

Adoption by a subset predicts most members unchanged with a right tail. Permeation predicts the whole distribution translating. Per-member shift, each against that member's own frequency-matched placebo, using the equal-weight statistic that carries the corpus effect:

chamber	median member	members shifting positive	top decile's share
Dáil Éireann	+0.089	78%	32%
UK Commons	+0.071	69%	33%
Canada Commons	+0.008	53%	47%

A statistic-choice warning. Measured as a *pooled* per-speaker rate this reverses — Canada 22% positive, UK 49% — because pooling is frequency-dominated while the effect lives in rare words. Same data, opposite conclusion. Use equal-weight.

Deleting the most-shifted members and re-running the frozen protocol:

corpus	baseline	–top 10%	–top 25%	–top 50%
Dáil Éireann	+0.246	+0.199	+0.166	+0.130
Canada Commons	+0.252	+0.155	+0.158	+0.147

About **half** the effect is carried by the top movers and about half survives deleting them, still at $p < 0.001$ with half the chamber gone. So neither story is complete: there is real concentrated adoption *and* a broad-based floor.

That test is biased and the bias mattered. Speakers were ranked by the same quantity then measured in the remainder — selection on the outcome, which lowers the trimmed mean even under pure permeation. The clean version ranks members on odd-numbered sittings and measures on even ones, so ranking and measurement use independent data:

corpus (measured on fold B)	baseline	–top 10%	–top 25%	–top 50%
Canada Commons	+0.351	+0.357	+0.314	+0.325
UK Commons	+0.317	+0.278	+0.236	+0.194
Dáil Éireann	+0.277	+0.233	+0.201	+0.165

All three survive deleting half the membership, at $p < 0.001$. Canada is **essentially unchanged** (7% reduction, against 42% in the biased version — so that 42% was almost entirely the artifact). Ireland and the UK fall about 40% but remain strongly positive. **No minority of adopters can account for the effect in any chamber.**

5.2c Unscripted business — and the result that cuts the other way

Length is only a proxy for spontaneity. Parliamentary procedure gives the real thing: business where the words cannot have been drafted, because nobody knew what was coming. Canada's oral questions are not submitted in advance at all; UK Topical Questions are not tabled and PMQs supplementaries are a surprise; Ireland's Leaders' Questions and Questions on Promised Legislation are raised without notice.

chamber	unscripted	prepared	ratio
Dáil Éireann	+0.362	+0.220	1.65×
UK Commons	+0.289	+0.261	1.10×
Canada Commons	+0.287	+0.405	0.71×

Canada reverses it, and Canada is the sharpest test. For the UK and Ireland the "prepared" pole is just *everything else* — a diluted mixture. In Canada both poles are clean categories: Oral Questions against **Statements by Members**, the 60-second set-pieces that are the most scripted text in the chamber. There, the shift is largest exactly where drafting is easiest.

Ordering the Canadian strata by how scriptable they are makes the point:

Canadian stratum	excess
Statements by Members (fully scripted)	+0.405
all short segments	+0.312
Oral Questions (unscripted)	+0.287
all long segments	+0.221

The purer the prepared category, the larger the shift — the long-band average is low only because ordinary debate dilutes it. The earlier length-band reading of Canada was an artifact of that dilution.

5.2d What the permeation arm actually supports

Both mechanisms are real, and their relative weight differs by chamber in a way that **tracks measured AI prevalence**:

chamber	AI prevalence	unscripted/prepared	reading
UK Commons	0.8%	1.10×	mostly permeation
Dáil Éireann	4.3%	1.65×	mostly permeation
Canada Commons	10.6%	0.71×	drafting visible on top of it

That coherence is itself evidence the two instruments are measuring what we think. But the honest headline is **not** "the shift is permeation". It is:

1. **A broad-based floor exists everywhere.** It survives deleting half of each chamber's members under an unbiased split-half design, and it appears in speech that could not have been drafted. Something is changing in how people write and speak who are not using these tools on that text.
2. **Direct drafting sits on top of it, and is chamber-specific.** Where prevalence is highest, the shift concentrates in scripted formats.

Claim 1 is the mass-behaviour-change result and it is supported. Claim 2 is not a threat to it — they are separable, and separating them is what these tests do.

5.2d-bis In-context permeation — the phrasing, not just the words (Matthew's design)

Frequency counting shows members using more of the register vocabulary. A sharper question is whether, when they use those words, the *surrounding phrasing* has become the assistant's phrasing — collocations and frames, not lexicon. Measured as the per-token log-likelihood advantage of an instruct checkpoint over its own base checkpoint, **at instrument-word positions only**, self-normalised by each segment's full-trace delta (no placebo word list — that construction has failed three times in this study and is not used here).

Pre → post change in that quantity, two model families, five chambers:

family	ie	ca	uk	ush	uss	pooled
Qwen3-8B	+0.017	+0.016	+0.011	+0.013	+0.002	+0.0118
Mistral-7B	+0.005	+0.008	+0.017	−0.006	+0.016	+0.0081

Two-family pooled: +0.0099, 95% CI [+0.0007, +0.0196], $P(\leq 0) = 0.020$; 9 of 10 family-chamber cells positive (clustered bootstrap over speeches).

This is the first *contextual* permeation evidence: the phrasing around the register words has drifted measurably toward the assistant's, in text selected identically in both eras. It is detector-free (the humanizer objection does not apply) and word-selection-free (self-normalised). The magnitude is small — about a hundredth of a nat per token — and the corpus- wide version of the same quantity is a triple-replicated null, so the drift is confined to the neighbourhoods of the register vocabulary, which is exactly where a diffusion-of-phrasing account puts it.

5.2e Shift of the centre, or collapse toward it?

Everything above is a **mean**. A mean can move two ways with opposite implications: everyone moves together and the distribution *translates*, or members converge on a common register and the distribution *narrows*. The second is the monoculture claim — individual voices becoming interchangeable — and it is a statement about second moments, not means.

Every member is subsampled to **exactly 10,000 tokens in both eras**, and only members present in both are used. Without that, the smaller post window would manufacture convergence by itself, since every diversity and distance measure is biased by sample size.

measure	Ireland	Canada	UK
instrument rate (the mean)	+90	-271	-44
between-speaker SD	+42	+89	-5 (<i>n.s.</i>)
Burrows Delta between members	+0.079	+0.079	+0.042
Shannon entropy	+0.026	-0.071	+0.029
distinct types	-288 (-1.2%)	-1710 (-6.9%)	-467 (-1.4%)
top-100 token share	-0.005	+0.005	-0.004

All significant at 95% except the UK's between-speaker SD.

On the measure that directly tests the question, it is a shift of the centre, not a collapse. Burrows Delta — the standard authorship-distance statistic, z-scored profiles over the 200 most frequent words — **rises in all three chambers**. Members became *more* distinguishable from one another, not less. Between-speaker dispersion of the instrument rate also rises in two chambers and is flat in the third. Nothing is converging on a common level of use.

Delta is a *pairwise* distance, so a shift affecting everyone equally cancels out of it exactly. A rise therefore means genuine added heterogeneity, and it cannot be an artifact of the common drift the rest of this study measures.

But one thing does narrow, everywhere: the size of the working vocabulary. Distinct types fall in all three chambers at fixed token count, and in Canada the corpus also concentrates outright — entropy down, top-100 share up. So the picture is:

- **individual distinctiveness: rising** — no collapse of voice
- **collective lexicon: shrinking** — fewer distinct words in use

Canada shows both at once, which is not a contradiction: Delta is computed on function-word-dominated profiles and is topic-robust, so its members can diverge stylistically while their shared topic vocabulary narrows.

The honest caveat. Type-count narrowing has an obvious non-AI explanation, and Canada — where it is largest by five-fold — is precisely the chamber with documented massive topic churn (bail 17.9×, corruption 6.2×, auditor 4.7×). A narrower political agenda produces a narrower lexicon with no machine involved. Attributing the vocabulary contraction to AI is not yet supported; attributing the *absence of stylistic collapse* to the data is.

And a limit on what a two-window design can see. Early adoption should *add* variance — some members pick it up, others don't — so widening is what the early phase of a convergence process looks like from the outside. Collapse would be a late signature, after everyone has converged on the same tool. This does not refute the monoculture prediction; it dates it. Between-speaker Delta is the series to track over time, and a single pre/post contrast cannot substitute for that.

6. The p-hacking question, handled explicitly

Three defences, in increasing order of strength:

1. **The instrument is external and predates the study.**
2. **The primary statistic involves no choices** — no threshold, no subsetting, all 407 words, equal weight.

An earlier version of this defence claimed a rare-word threshold sweep "shows every cutoff from <1 to <50 per 100k gives $p \leq 0.008$ ". That was New Brunswick only, unlabelled, and misquoted (NB's <1 cell is 0.015). **The frequency-weighted secondary does not replicate**, and the full table is now reported rather than one chamber's:

chamber	<1	<2	<5	<10	<20	<50
New Brunswick (<i>discovery</i>)	0.015	0.008	0.000	0.001	0.000	0.000
US House	0.020	0.023	0.020	0.062	0.213	0.603
Dáil Éireann	0.207	0.146	0.083	0.047	0.069	0.136
UK Commons	0.143	0.184	0.148	0.240	0.130	0.111
Canada Commons	0.137	0.332	0.195	0.274	0.489	0.916
US Senate	0.472	0.945	0.984	0.964	0.986	0.992

4 of 30 confirmatory cells reach $p < 0.05$. The pooled statistic is frequency-weighted, so a handful of common words dominate it — §2.2 shows it is null even in NB on the full list (ratio 1.09, $p = 0.48$) while the equal-weight statistic is +0.379. The choice of equal weighting is defended on the merits and follows Gray's published design; what is *not* defensible is letting a reader assume the two statistics agree. They do not. 3. ~~~~A mechanism prediction test.~~~~
REMOVED, along with the corpus it depended on. It claimed that word-level AI-preference — measured from 40 synthetic speeches and the 2019 control — predicts which individual words rose (Spearman +0.548, $p < 10^{-4}$). Two things killed it, and the second subsumes the first.

The statistic was a rarity index. The 2019 control is a subset of the 2018–22 pre-window, so each word's Hansard rarity sat inside both the predictor and the outcome with the same sign. It correlates **+0.923** with $-\log(2019 \text{ count})$ and **-0.234** with the synthetic evidence it was meant to encode; the partial correlation controlling pre-period frequency is **-0.008**; and permuting within frequency quintiles rather than globally gives **$p = 0.24$** . Rarity alone, using no AI data at all, scores **+0.568** — better than the published predictor.

The synthetic corpus is gone. Its 6,838 words were too few to carry variance (39 of the 89 usable words occurred exactly once), and it came from a single 7B open-weight model that does not represent what anyone actually drafts with. Nothing in the study now depends on it.

So this is not a defence with a correction pending — the analysis has been deleted. What is retained is its *design lesson*: an unpaired comparison of model output against a human corpus inherits that corpus's word-frequency structure on both sides. The replacement (§6.1) is paired by construction — instruct against its **own base checkpoint** on identical prompts — which cancels baseline frequency exactly, as in Geng & Trotta and Yakura et al. Defences 1 and 2 carry §6 in the meantime.

5.4 Where the register comes from — onset, cohort, and arrival

The lexicon arm failed as an AI instrument (§2.6b) but the series it measures is real and turns out to have structure worth reporting in its own right. Three findings, from a 16-chamber panel built for this purpose: eight Canadian provinces, six Australian states, and the three UK devolved legislatures, each from its own archive with its own extractor.

5.4.1 The onset is 1994–96, not 2022

Extending the UK Commons series to **1985–2026** (10,006 sitting-day files) shows the register *declining* before the consumer web and rising after:

period	slope (gap units/yr)
1985–1995	-18.97
1996–2005	+31.64
2006–2016	+42.22
2017–2026	+28.26

The gap falls from 774 (1985) to a minimum of **593 (1994)**, then climbs monotonically to 1,720 (2026). The frequency- and-dispersion-matched placebo is flat across all 41 years (1,520–1,575), so this is not measurement drift. A pre-existing secular trend would have been rising in 1985–94; it was falling. **The inflection coincides with mass consumer internet adoption, and the transformer era is the slowest-growing of the three rising periods.**

5.4.2 It arrives with new members, not with changed ones

quantity (16 chambers)	pooled	95% CI
arrival premium (new members minus incumbents, same years)	+1.87	[+1.25, +2.49]
within-member change	−0.42	[−1.32, +0.49] (<i>n.s.</i>)
compositional component	+2.12	—

Positive in **15 of 16 chambers** (sign test $p \approx 2.6 \times 10^{-4}$), significant individually in 10. Member-level bootstrap, resampling members not words.

The honest form of this is "**incumbents flat, arrivals higher**" — not "incumbents declined". The within-member component is significantly negative in only 4 chambers (BC, ON, QLD, SK) and significantly *positive* in 3 (NI, SCO, WAL). Those three are all UK devolved, and two carry known era-asymmetric instruments (Wales PDF → XML across the windows; NI's second window truncated by the Stormont collapse), which is exactly the artifact shape that manufactures a spurious within-member effect.

5.4.3 What distinguishes arrivals is birth year

Per-member-year register rate on birth decade, province and year fixed (3,445 member-years, 79% birth-year coverage from Wikidata/Wikipedia):

predictor	coefficient	t
birth decade	+0.93	+12.03
birth decade, adding occupation + education	+1.06	+13.36
post-secondary attested	−0.53	−3.31
occupation: law	−1.67	−5.83
occupation: communications/PR/journalism	−0.67	−2.02

Raw cohort means run 33.30 (1930s) → 33.64 (1950s) → 36.84 (1970s) → 39.60 (1990s) — flat, then climbing from the 1960s cohort.

Candidate professionalisation and educational expansion are refuted as explanations: communications backgrounds use *less* of the register, as do law and the post-secondary-educated, and controlling for both *strengthens* the cohort coefficient.

Cohort, not age. With year fixed effects, birth year and age are collinear, so the coefficient could in principle be an age effect. It is not: mean chamber age is flat (52.2 in 2006 → 51.4 in 2019) while the register rises +2.06. A stable-age chamber shows no trend under an age effect. Under a cohort effect, mean birth year advanced 13.8 years, which at +0.093/yr predicts **+1.28 of the observed +2.06 — about 60%.**

5.4.4 What is still not identified — three nulls

Birth cohort is a portmanteau for everything that differs between generations. Three attempts to isolate computer exposure specifically have now returned null, each fixing the previous one's flaw:

attempt	design	result
1	district adoption during tenure (US states; provinces)	null; -0.008 ($t = -0.64$) and -0.0045 [$-0.0108, +0.0009$]
2	cohort $\times$ service -province adoption	$+0.09$ ($t = +0.97$)
3	cohort $\times$ birth -province adoption, formative window	see below

Attempt 3 used birthplace collected from Wikidata P19 (70% coverage) — the right geography at last, since **34% of members serve a province other than their birth province**, half of them foreign-born across 34 countries. Three specifications, pre-committed in that order:

specification	estimate	verdict
A — direct: birth-province adoption during the member's own age-15–25 window	-0.40 ($t = -1.00$)	null, wrong sign, 106 people
B — interaction , all cohorts	$+0.21$	$t = +2.20$ on member-year SEs, but CI [$-0.22, +0.33$] clustered on the 10 birth provinces
C — birth-settlement size (urbanicity)	$+0.15$ ($t = +1.37$)	null

B is a cautionary case worth keeping in the record. Its regressor varies only across birth provinces, so member-year standard errors treat 2,394 observations as independent when there are ten clusters. Correctly clustered, 23% of resamples fall at or below zero. This is the second time in this study that unclustered inference on a province-level regressor produced a spurious "significant" result, and both times it flattered the hypothesis.

Meanwhile **the cohort effect is unmoved by any of this:** birth decade $+1.02$ to $+1.22$, $t \approx +11$ – 12 , in every specification.

So: the register enters legislatures through generational replacement, and no measure of technology exposure we can construct predicts which members bring more of it. The defensible claim is **generational, mechanism unidentified**. Ambient computing remains the leading candidate on **timing** — the onset at 1994–96 in §5.4.1 — but that is a coincidence of dates, not a dose-response, and three dose-response tests have now failed. Further proxies are not worth running; the informative next evidence would be a different kind, not a fourth operationalisation of the same one.

5.3 The exposure-gradient tests — two nulls, and a continental replication

If ambient computer immersion drives the register drift (the strongest causal reading of its ~ 2006 onset), drift intensity should gradient on local adoption. Two designs tested this, both entirely inside pre-LLM windows (member drift 2006–2010 $\rightarrow$ 2015–2019):

US Congress — drift vs home-state internet adoption (NTIA Sept-2001 CPS): coefficient -0.008 per sd ($t = -0.64$), null, with demonstrated power (the same regression finds Republicans drifting $+0.093$, $t = +3.40$). Escape hatch: congressional drafting staff live in Washington, so home-state adoption may never reach the drafting environment.

Canadian provinces — the version without the escape hatch: provincial staff live in-province, offices are small, many members self-draft. Exposure from StatCan 22-10-0034 (household internet use, by province; 2000 spread 43.6–58.8). Seven chambers built from their own Hansard archives, 259 name-resolved members with $\geq 5k$ words in both windows:

province	adoption 2000	mean drift
Alberta	58.8	+0.002
BC	55.9	+0.042
Ontario	54.2	+0.061
Nova Scotia	52.0	+0.082
Manitoba	49.8	+0.094
Saskatchewan	46.9	+0.074
Newfoundland	45.5	+0.062

Registered spec (2000 level): slope **-0.0045**, province-clustered bootstrap CI [-0.0108, +0.0009] — wrong sign for exposure. Alternative spec (adoption growth 2000 → 2003, the saturation-lag reading): **+0.0059** [-0.0062, +0.0153] — null, and the growth column visibly fails to order the drift.

Verdict: the register climb does not dose-respond on local computer adoption at either margin, in either country. The exposure mechanism in its geographically testable form is unsupported; cultural accounts that operate nationally (or continentally) remain open, but they are exactly the accounts a geographic gradient cannot test.

The finding that emerged anyway: six of seven provinces show strong pre-LLM register drift (+0.04 to +0.09 over nine years). The UK's secular climb replicates across seven independent chambers with independent Hansard offices. The register change is continental and predates the tools everywhere examined.

Caveats: seven clusters; Alberta is an outlier (+0.002, $n = 19$, PDF-era extraction); and each province passes through a bespoke extractor with markup-era boundaries inside the design windows (BC 2008, ON 2011, NS 2010), so era-asymmetric extraction artifacts load directly on the cross-province coefficient. The NULL is robust to this — no artifact pattern plausibly creates an absent positive gradient — but the negative point estimate should not be quoted without an era-boundary robustness pass.

6.1 The alignment-register experiment — one result, one failure

A paired design was built to replace the withdrawn test and to ask a separate question: is the register that leaked specifically the **alignment** register? An instruct model differs from its own base model only by post-training, so what SFT-plus-preference-optimisation adds is directly measurable. Identical prompts (openings of real pre-2023 segments), identical decoding and seeds, no chat template — so the only difference is the weights. Two families completed: Qwen3-8B and Mistral-7B. Generations where the instruct model *analysed* the prompt rather than continuing it are dropped from **both** members of the pair, since that is instruction-following behaviour rather than register.

The result that holds — Kobak's instrument is substantially an alignment artifact. Mean $\log(\text{instruct}/\text{base})$ for the style words, against frequency-matched controls drawn from the base model's own output:

family	excess	95% CI	prompt pairs
Qwen3-8B	+1.04	[+1.05, +1.31]	668
Mistral-7B	+0.65	[+0.67, +0.99]	788

Pooled excess **+0.880**, $p < 0.001$, and it is broad rather than a few words: **70.8% of style words are instruct-preferred against 28.2% of all other words.** So the vocabulary Kobak derived from PubMed abstracts — with no reference to alignment at all — is largely the vocabulary that post-training adds. "AI register" is more precisely **assistant register**.

The test that fails — it does not predict Hansard. The same score, asked whether it predicts *which words rose*, does not survive frequency stratification:

predictor	Spearman	partial freq	stratified p
alignment preference (paired)	+0.112	+0.097	0.098
generic model-vs-human (unpaired)	+0.044	+0.006	0.581
— Qwen3 alone	+0.045	+0.034	0.314
— Mistral alone	+0.012	+0.031	0.298

Neither family replicates individually and the pooled value misses 0.05. **The replacement for the withdrawn mechanism test is not delivered.** (An earlier run gave $p = 0.031$; that was before the meta-commentary filter, and the filtered number is the honest one.) The rarity check does confirm the design works as intended — the new predictor scores +0.22 against log-rarity where the dead one scored +0.92 — so the failure is a real null, not a repeat of the same confound.

What post-training actually adds (descriptive, mean log instruct/base):

category	value
discourse structuring — <i>moreover, furthermore, however, ultimately</i>	+0.43
significance marking — <i>important, crucial, vital, essential</i>	+0.33
constructive verbs — <i>ensure, foster, enhance, promote</i>	+0.16
hedging — <i>perhaps, arguably, somewhat, potentially</i>	-0.02

This refines the sycophancy hypothesis rather than confirming it. What preference tuning adds is **scaffolding and evaluative inflation**, not hedging or politeness — hedges are flat to slightly *down*. If the assistant register is leaking into parliamentary speech, the leak is "moreover" and "crucial", not "perhaps".

Limits. Two families, not three — Llama-3-8B-Instruct has no local weights and is gated. The two disagree in magnitude by $\sim 1.6\times$. The descriptive list still carries some task residue (*text, speaker, context*) because the filter only inspects each generation's opening. And an instruct-model contrast is not the same as measuring RLHF specifically: SFT and preference optimisation are confounded here. Separating them needs a family that ships intermediate checkpoints — OLMo-2 publishes SFT and DPO stages separately and is the right next step.

And the honest concession: **New Brunswick was selected because a signal was visible there** (the origin was a clip of a member reading an AI framing sentence aloud). That selects on the outcome even though the instrument is external. NB is therefore labelled the **discovery** corpus, quoted as an upper bound, and excluded from the Fisher combination.

7. Quality measurement

Instrument: the **Discourse Quality Index** (Steenbergen, Bächtiger, Spörndli & Steiner 2003) — the standard deliberation-quality scale for parliamentary debate, whose own worked examples come from a 1998 Commons debate. Ten of those published codings are embedded as in-context anchors, so the scale is the authors', not ours.

Two traps designed against. **Circularity:** most "quality" metrics are style metrics, and AI text differs in style by construction, so every measure carries a circularity grade and the judge separately reports P(AI-assisted) purely to detect leakage. **Genre:** AI-flagged segments concentrate in prepared statements, so comparisons run within selection-matched pools.

Reliability: repeat-pass Spearman 0.68–0.91 per dimension, at or above the published human inter-coder bar (their justification $r = 0.716$).

Findings: AI-flagged speech scores **higher** on formal justification, ~3× **lower** on first-person witness, and **lower** on engaging opponents' demands. Blinded 2019 → 2026: justification +0.21 and evidence +0.19 up, **respect toward others' demands −0.36 (0.96 → 0.60)** and constructive politics −0.30 (0.54 → 0.24) down. Summary: **form up, deliberation down**.

(An earlier draft of this section gave the respect figure as 0.96 → 0.52. The anchored-v2 run's own record, `quality_methods.md`, says 0.60; 0.52 was a transcription error and the corrected value is above.)

Genre remains the unsolved confound, and recent work makes it sharper. The design compares within selection-matched Pangram strata, but full prepared-vs-spontaneous labels were always an open item. We now know from the unscripted-business analysis (§5.2c) that AI concentrates in scripted formats — Canada's Statements by Members carry a larger shift than Oral Questions. So "AI-flagged speech engages opponents less" may in part be "scripted speech engages opponents less". The provincial corpora built since carry business- rubric metadata that would let this be settled rather than stated; it has not been done.

The pronoun result connects to a real literature — Pennebaker and colleagues found high-status speakers use fewer first-person singulars, and deception research found the same for fabricated accounts. Here it is neither: it is **authorship displacement**, because the drafting voice has no experience to reference. A corollary worth stating: pronoun-based psychological inference on post-2023 institutional text is now unsafe.

8. What each headline number licenses you to say

number	claim it supports	claim it does NOT support
$p = 2.4 \times 10^{-7}$ (Fisher, 3 chambers)	the vocabulary shift is real and replicable	anything about <i>how much</i> text is AI
excess +0.218 to +0.272	consistent magnitude across parliaments	that it is large
8.3% prevalence (NB)	~1 in 12 recent NB segments is AI-flagged	anything about other chambers (<i>pending</i>)
Sp = 1.0 measured (423/423)	no inflation from false positives	that Se = 1 on <i>edited</i> AI — unmeasured, so 7.5% is a floor (§4.3)
89.6% of 643 confirmed	the screen is a precise stratifier	a corpus-wide rate
AUC 0.951 (Opus screen)	frontier LLMs track a commercial detector	that either tracks ground truth

And one wording constraint. In Canada the absolute volume of AI-style vocabulary *fell* (bootstrap CIs 0.925 and 0.916). It fell less than comparable vocabulary, which is what the significant result measures. The defensible phrasing is **"rises relative to comparable vocabulary"** — not "rises". That holds in all four chambers; the stronger claim does not.

9. Reproducing any of it

<code>run_protocol.py</code>	<code>CORPUS_NAME OUT_PREFIX segments.jsonl</code>	# the frozen protocol
<code>formality_axis.py</code>	<code>CORPUS_NAME segments.jsonl</code>	# data-defined control
<code>cross_corpus.py</code>		# table + Fisher + figure

The protocol's only per-corpus input is the corpus name, which seeds the RNG via `sha1(name)[:8]`. Per-corpus code exists solely to map a source format onto the shared segment schema. That separation is what makes the cross-chamber combination legitimate rather than a fishing expedition.